

The axioms of n -exact categories

BACHELOR'S THESIS

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Abstract

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Introduction

0.1 Prerequisites

In order to understand this paper, the reader should have some knowledge about modern homological algebra. In particular, they should already be familiar with category theory, more specifically abelian category theory, as is presented for example in the lecture notes of MA3204 Homological Algebra [\[2\]](#)

Conventions and notations

Chapter 1

An Introduction to Exact Categories

As a way to generalize the notion of abelian categories, one can look at a general additive category on which we define a certain notion of short exact sequences that has very similar properties to the abelian category counterpart. This chapter aims to motivate the definition of these so-called exact categories, to later define them formally and generalize some classical results from abelian categories.

In this chapter, the word kernel (respectively cokernel) will usually refer to the kernel morphism (respectively cokernel morphism), not the object. The content of this chapter is inspired by both the lecture notes of MA3204 Homological algebra [2] and the article about exact categories by Theo Bühler [1].

1.1 Motivation

When studying abelian categories, there are many convenient properties related to monics and epics. Let us recall a few:

Proposition 1.1 *In an abelian category, the following properties hold:*

- *the pushout of a monic is a monic;*
- *the pullback of a monic is a monic;*
- *the pushout of an epic is an epic;*
- *the pullback of an epic is an epic.*

Another nice property is the following, which further illustrates the behavior of monics and epics in abelian categories: This property is particularly useful because it allows us to understand the structure of morphisms in abelian categories in terms of kernels and cokernels. Specifically, it tells us that monics and epics can be characterized by their relationship to kernels and cokernels, respectively. This characterization is crucial for many proofs and constructions in category theory, as it provides a way to decompose and analyze morphisms using well-understood concepts. Additionally, this property highlights the symmetry between monics and epics in abelian categories, which is a recurring theme in the study of exact sequences and homological algebra.

Proposition 1.2 *In an abelian category,*

- *every monic is a kernel of its cokernel;*
- *every epic is a cokernel of its kernel.*

This allows us to conclude many properties about kernels and cokernels:

Proposition 1.3 *Let $\mathcal{A}$ be an abelian category and let $\mathcal{E}$ be the class of short exact sequences. $\mathcal{E}$ is closed under isomorphism. That is, if*

$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ is a short exact sequence and we can form the commutative diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C \longrightarrow 0 \\
 & & \downarrow a & & \downarrow b & & \downarrow c \\
 0 & \longrightarrow & A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \longrightarrow 0
 \end{array}$$

where the vertical arrows are isomorphisms, then $0 \rightarrow A' \xrightarrow{f'} B' \xrightarrow{g'} C' \rightarrow 0$ is a short exact sequence.

Furthermore, the following hold:

- [E0] For any object $A \in \mathcal{A}$, the identity morphism 1_A is a kernel;
- [E0^{op}] For any object $A \in \mathcal{A}$, the identity morphism 1_A is a cokernel;
- [E1] The class of kernels is closed under composition;
- [E1^{op}] The class of cokernels is closed under composition;
- [E2] The pushout of a kernel along any morphism is a kernel;
- [E2^{op}] The pullback of a cokernel along any morphism is a cokernel.

Proof It is clear that the class of short exact sequences is closed under isomorphism, since. Indeed, $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ is a short exact sequence if and only if f is a kernel of g and g is a cokernel of f . The result follows from the fact that kernels and cokernels are preserved by isomorphisms.

[E0] and [E0^{op}] follow directly from the fact that the identity morphism is both the kernel and cokernel of the zero map.

We can deduce [E1] and [E1^{op}] by recalling that composing monics (respectively epics) gives us monics and that monics are kernels of their cokernel (respectively epics are cokernels of their kernel).

As for [E2], recall that kernels are always monic. Thus, by 1.1, the pushout of a kernel is monic as well. But then, by 1.2, it is a kernel of its cokernel, thus it is a kernel. The proof for $[E2^{\text{op}}]$ is similar. $\square$

1.2 Exact Categories

Based on the observations from the last section, one can define a more general notion that encapsulates the idea of short exact sequences, namely exact categories. To do so, let us first define kernel-cokernel pairs.

Definition 1.5 Let $\mathcal{A}$ be an additive category. A *kernel-cokernel pair* (i, p) in $\mathcal{A}$ is a pair of composable morphisms $A \xrightarrow{i} B \xrightarrow{p} C$ such that i is a kernel of p and p is a cokernel of i .

In the context of abelian categories, we notice that kernel-cokernel pairs indeed give rise to short exact sequences in the usual sense. In a general additive category, however, we cannot say the same since we have no guarantee that kernels and cokernels even exist for arbitrary morphisms, and thus we are unsure that $\text{Im } i$ exists. But we will use our intuition from abelian categories to expand the notion of short exact sequences to additive categories. First, let us recall what it means for a class to be closed under isomorphism.

Definition 1.6 Let $\mathcal{A}$ be an additive category and $\mathcal{E}$ be a class of kernel-cokernel pairs in $\mathcal{A}$. We call a morphism $A \xrightarrow{i} B$ an *admissible monic* (or *inflation*) if there exists a morphism $B \xrightarrow{p} C$ such that $(i, p) \in \mathcal{E}$. Dually, we call a morphism $B \xrightarrow{q} C$ an *admissible epic* (or *deflation*) if there exists a morphism $A \xrightarrow{j} B$ such that $(j, q) \in \mathcal{E}$. We use $A \xrightarrow{i} B$ for admissible monics, and $B \xrightarrow{p} C$ for admissible epics. We say $\mathcal{E}$ is closed under isomorphism if whenever we have $A \xrightarrow{i} B \xrightarrow{p} C \in \mathcal{E}$ and the following commutative diagram

$$\begin{array}{ccccc}
 A & \xrightarrow{i} & B & \xrightarrow{p} & C \\
 \downarrow a & & \downarrow b & & \downarrow c \\
 A' & \xrightarrow{j} & B' & \xrightarrow{q} & C'
 \end{array}$$

with a , b , and c isomorphisms, then $(j, q) \in \mathcal{E}$.

It is usually convenient to require being closed under isomorphism, as objects with universal properties are usually unique up to isomorphism. In particular, recall that all kernel objects of a specific morphism are isomorphic. In the case of a class of kernel-cokernel pairs $\mathcal{E}$, being closed under isomorphism implies that all kernels of an admissible epic are admissible monics. Indeed, consider $B \xrightarrow{p} C$ an admissible epic and $A \xrightarrow{i} B$ an admissible monic such that $(i, p) \in \mathcal{E}$. Let $A' \xrightarrow{j} B$ be another kernel of p and $A \xrightarrow{a} A'$ be the isomorphism between both kernels. We can construct the following commutative diagram

$$\begin{array}{ccccc}
 A & \xrightarrow{i} & B & \xrightarrow{p} & C \\
 \downarrow a & & \downarrow 1_B & & \downarrow 1_C \\
 A' & \xrightarrow{j} & B & \xrightarrow{p} & C
 \end{array}$$

Since all vertical arrows are isomorphisms, we can conclude that $(j, p) \in \mathcal{E}$. Dually, we can show that all cokernels of an admissible monic are admissible epics.

We can now proceed to define exact categories, basing our definition on the properties we established for abelian categories:

Definition 1.7 Let $\mathcal{A}$ be an additive category. An *exact structure* on $\mathcal{A}$ is a class $\mathcal{E}$ of kernel-cokernel pairs in $\mathcal{A}$ that is closed under isomorphism and satisfies the following axioms:

- [E0] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is an admissible monic;
- [E0^{op}] For all objects $A \in \mathcal{A}$, the identity morphism 1_A is an admissible epic;
- [E1] The composition of two admissible monics yields an admissible monic;
- [E1^{op}] The composition of two admissible epics yields an admissible epic;
- [E2] The pushout of an admissible monic along any morphism yields an admissible monic;
- [E2^{op}] The pullback of an admissible epic along any morphism yields an admissible epic.

The last two axioms can be summarized by the following commutative diagrams:

$$\begin{array}{ccc}
 C & \xrightarrow{f'} & D \\
 \downarrow c & \text{PO} & \downarrow d \\
 C' & \xrightarrow{\quad f \quad} & D'
 \end{array}
 \quad \text{and} \quad
 \begin{array}{ccc}
 C' & \overset{\quad f' \quad}{\dashrightarrow} & D' \\
 \downarrow c & \text{PB} & \downarrow d \\
 C & \xrightarrow{\quad f \quad} & D
 \end{array}$$

If $\mathcal{E}$ is an exact structure on $\mathcal{A}$, we say $(\mathcal{A}, \mathcal{E})$ is an *exact category* and we call the elements of $\mathcal{E}$ *short exact sequences* (or *conflations*).

Chapter 2

Expanding to n -exact categories

Chapter 3

Generalizing results from
exact to n -exact categories

Bibliography

- [1] Theo Bühler. Exact categories. *Exposition. Math.*, 28(1):1–69, January 2010. [7](#)
- [2] Steffen Oppermann. Homological Algebra. https://wiki.math.ntnu.no/_media/ma3204/2021h/homalg.pdf. [3](#), [7](#)